AUTOMATION & QA DEVELOPER

TASK 2 – API AUTOMATION WORKFLOW

Project: Hacker News Top 5 Story Digest
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Platform: n8n **Timezone:** Asia/Kolkata

Student Project Report

1. Introduction

This project demonstrates how an automation workflow can collect information from a public API, process the response, apply a condition and send a notification. I selected the Hacker News API because it provides public story data that can be used without an API key.

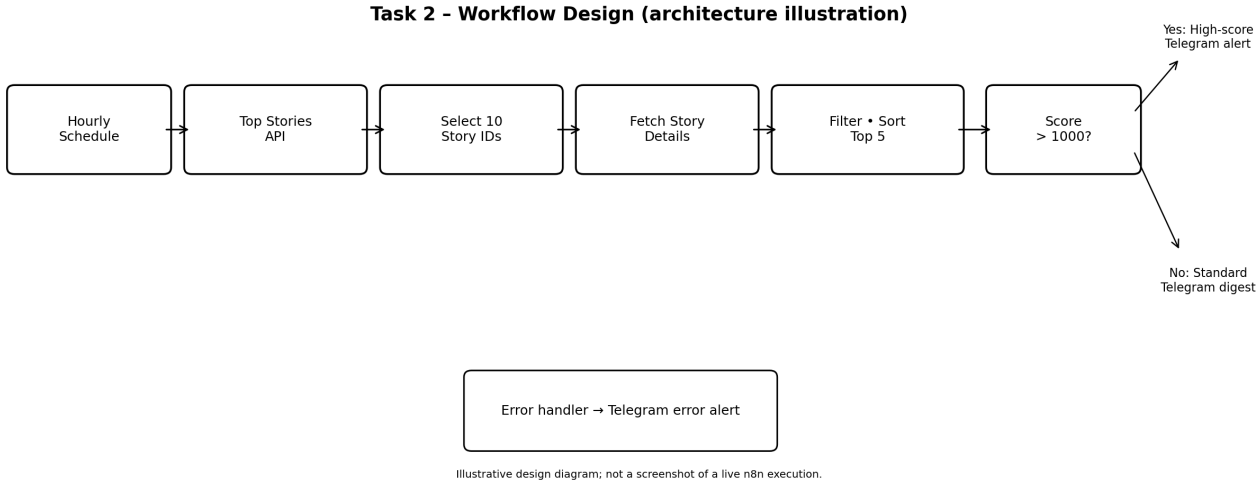
2. Objective

The objective is to create an hourly workflow that retrieves top Hacker News stories, fetches their details, filters and sorts them by score, selects the top five and sends the result to Telegram. A separate branch is included for stories with a score greater than 1,000.

3. Tools and APIs

Automation platform: n8n
Trigger: Schedule Trigger (every hour)
API 1: Hacker News top stories endpoint
API 2: Hacker News item details endpoint
Notification: Telegram Bot API through n8n Telegram node

4. Workflow Design



The diagram is an architecture illustration prepared for this report. It is not a screenshot from the n8n editor or a successful execution.

5. Workflow Explanation

Node	Purpose
Schedule Trigger	Starts the workflow once every hour.
Get Top Story IDs	Sends an HTTP GET request to the Hacker News top-stories endpoint.
Select First 10 IDs	Keeps the first ten IDs so the workflow makes a limited number of detail requests.
Fetch Story Details	Calls the item endpoint for each selected story ID.
Filter, Sort and Keep Top 5	Removes deleted, dead or incomplete story records, sorts the remaining records by score in descending order and keeps five.
IF Condition	Checks whether the story score is greater than 1,000.
Telegram Notification	Sends a high-score alert or a standard digest depending on the IF result.
Error Handling	Routes errors from the request and notification steps to an error message and Telegram alert.

6. Data Processing

The workflow uses the story title, URL, score, author, comment count and ID. The score is converted to a number before sorting. The final output is limited to five records. The IF node checks the numeric score against the threshold of 1,000.

7. Error Handling and Security

The HTTP Request and Telegram nodes are configured with error outputs so failures can be routed to an error handler. Telegram bot credentials are intended to be stored in n8n Credentials. The JSON file contains placeholder credential and chat ID values; no private token is included.

8. Test and Validation Plan

Test	Expected result	Status
Top-stories API	Returns an array of story IDs	Not run in live n8n
Story details API	Returns story data for selected IDs	Not run in live n8n
Filtering	Deleted/dead/incomplete records are excluded	Logic prepared; not live-tested
Sorting and limit	Highest scores first; maximum five records	Logic prepared; not live-tested
IF condition	Score > 1,000 uses high-score branch	Not run in live n8n
Telegram output	Digest arrives in configured chat	Pending credentials
Error branch	Failure notification is sent	Not run in live n8n

9. Implementation Status and Limitations

The workflow JSON has been prepared and validated as a JSON file. It has not been imported into a live n8n workspace or executed. Telegram credentials and the destination chat ID still need to be configured by the account owner. Because no live run has taken place, this report does not claim successful execution and does not include fabricated execution screenshots.

10. Conclusion

This project covers the main automation concepts required in the task: scheduled execution, two API requests, data transformation, filtering, sorting, top-five selection, conditional branching, notification delivery design and error routing. The next validation step is to configure Telegram credentials in n8n and run the workflow manually before enabling its hourly schedule.

References

Hacker News API: https://github.com/HackerNews/API
n8n Schedule Trigger: https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.scheduletrigger/

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